

**Design Thinking Project Report on
SMART HEALTH CARE APPOINTMENT SYSTEM**

Submitted to
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ABSTRACT

In an era where healthcare demands efficiency, traditional manual appointment systems falter with long wait times, double-bookings, and poor accessibility. This Smart Healthcare Appointment System addresses these challenges by delivering a web-based platform that digitizes appointment scheduling, management, and cancellation. Key features include real-time Live Queue Tracking for delay updates, Instant Notifications via email for reschedules or early slots, and a User-Friendly UI/UX with multi-language support and accessibility options. Built on a secure Node.js backend and MongoDB database, it reduces staff workload, enhances patient experience, and sets the stage for future scalability. Managing appointments in hospitals and clinics is often a slow and frustrating process for both patients and staff. This project focuses on solving those problems through a Smart Healthcare Appointment System—a web platform that makes booking, rescheduling, and tracking appointments much easier. It gives patients live updates on their queue status and sends automatic notifications for changes.

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1. INTRODUCTION

Background:

Healthcare globally struggles with outdated appointment systems—manual bookings and phone-based scheduling cause delays (often 30-60 minutes), frequent errors (e.g., double-bookings), and no centralized records, frustrating patients and overworking staff.

Importance of Digital Transformation:

Digitization streamlines operations, improves patient access, and supports data-driven decisions, aligning with global healthcare trends toward technology adoption.

Why Appointments? Why Now?

Appointments are the entry point to care, and with rising patient volumes and tech-savvy expectations, a modern solution is urgent. The post-pandemic shift to digital services underscores this need in 2025.

1.1 EXISTING SYSTEM

Manual Appointment Booking: Depends on paper or basic software, causing errors and taking up to 2 hours of staff time daily.

Phone-Based Scheduling: Involves long hold times, miscommunication, and no real-time updates, often resulting in overbooking.

Problems:

1. **Time-Consuming:** Delays patients and burdens staff.
2. **Error-Prone:** Double-bookings and lost records occur regularly.
3. **No Transparency:** Patients lack insight into wait times or slot status.
4. **No Record History:** Limits data analysis for improvements.

1.2 PROPOSED SYSTEM

Core Functionality:

A web platform where patients register/login, book appointments via a form (name, phone, email, hospital, doctor, time slot, date, reason), and check slot availability in real-time.

Innovative Features:

- **Real-Time Availability Check:** Prevents double-bookings dynamically via backend API.
- **User-Friendly UI/UX:** Consistent, modern design across signup, login, appointment, and about pages.
- **Session-Based Authentication:** Secure login/signup with user session tracking using Express sessions.

Implementation Details:

- **Backend:** Node.js with Express for API handling.
- **Database:** MongoDB for storing user credentials and appointment data.
- **Error Handling:** Clear alerts for issues (e.g., "Invalid credentials" or "Slot unavailable")

2. SYSTEM ANALYSIS

2.1 SOFTWARE REQUIREMENTS

Operating System:

- Windows 10 or later
- macOS 10.15 or later
- Linux distributions with kernel 4.15 or later

Software:

- Node.js (v18 or later)
- MongoDB
- Updated web browser (Chrome, Firefox, Edge)

Network:

- Stable internet connection for server and client communication

2.2 HARDWARE REQUIREMENTS

Processor:

- Minimum: Dual-core CPU (2.0 GHz or faster)
- Recommended: Quad-core CPU (2.5 GHz or faster)

Memory (RAM):

- Minimum: 4 GB
- Recommended: 8 GB or more

Storage:

- Minimum: 500 MB of available space for application files and database
- Recommended: 2 GB or more, including space for logs and future data growth

Display:

- Minimum: 10-inch screen with 1280x720 resolution
- Recommended: 13-inch screen or larger with 1920x1080 resolution or higher

Network:

- Minimum: Stable internet connection with 5 Mbps download/upload speed
- Recommended: 10 Mbps or higher for real-time performance

Server Hardware (for Backend):

- Minimum: 2-core CPU, 4 GB RAM, 10 GB SSD storage
- Recommended: 4-core CPU, 8 GB RAM, 20 GB SSD storage

3. SYSTEM DESIGN

3.1 ARCHITECTURE DESIGN

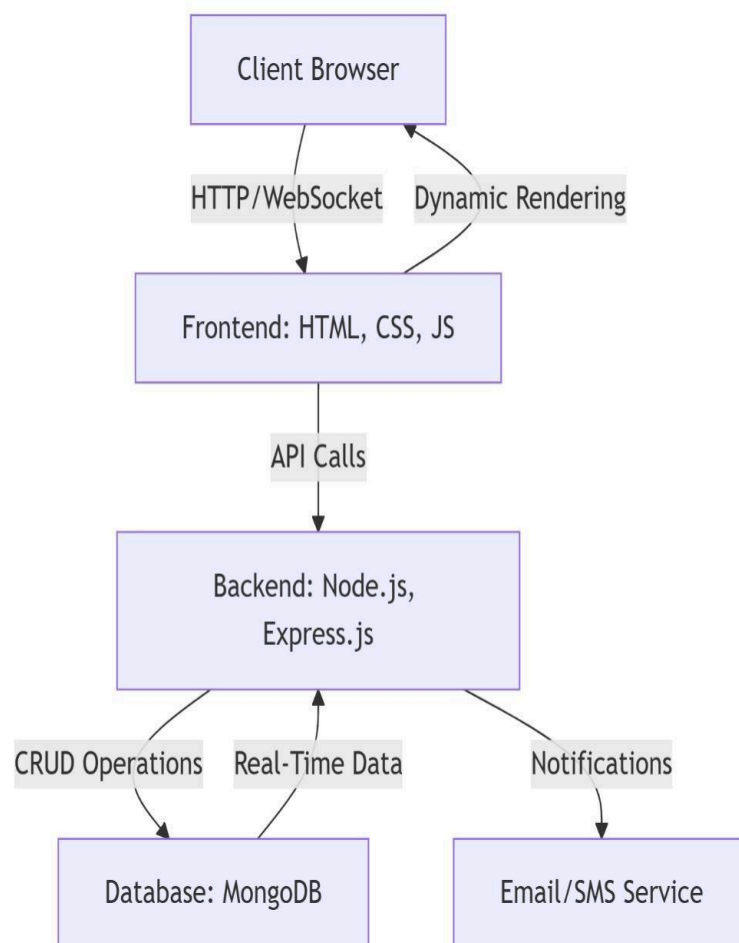


Fig 3.1 Architectural Design

3.2 FLOW CHART

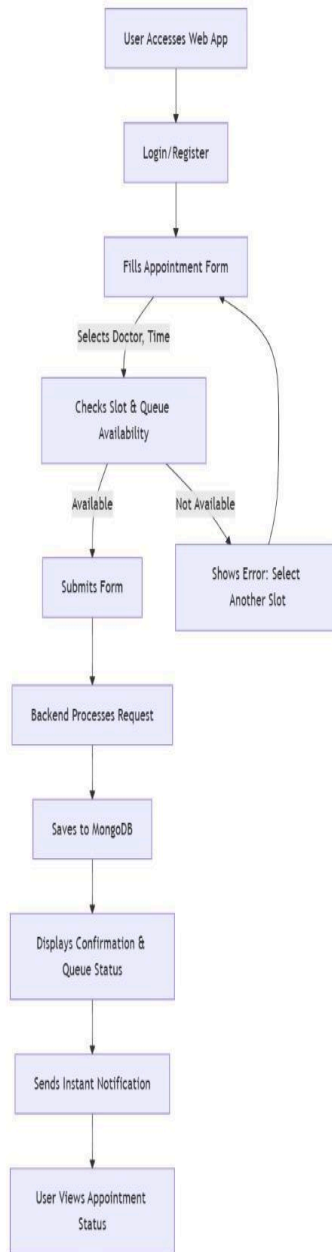


Fig 3.2 Flow Chart

3.3 MINDMAP DIAGRAM

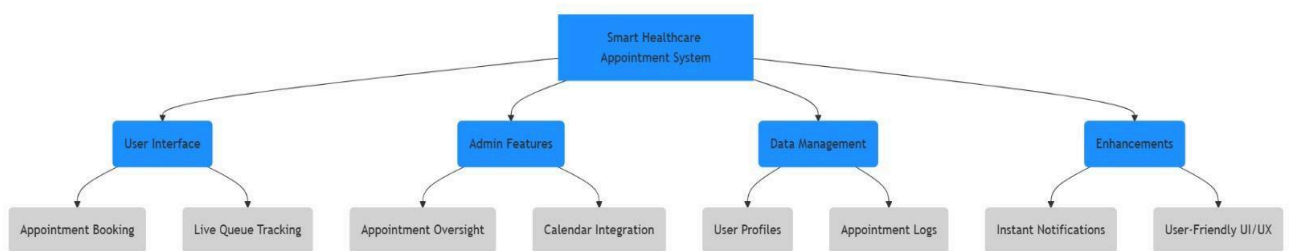


Fig 3.3 MindMap Diagram

4. CODE IMPLEMENTATION

Server.js

```
const express = require('express');
const path = require('path');
const mongoose = require('mongoose');
const session = require('express-session');
const app = express();
const PORT = 3000;

app.use(express.json());
app.use(express.urlencoded({ extended: true }));
app.use(express.static(path.join(__dirname, '../frontend')));
app.use(session({
  secret: 'your-secret-key',
  resave: false,
  saveUninitialized: false
}));

mongoose.connect('mongodb://localhost:27017/smarthealthcare')
  .then(() => console.log('✅ MongoDB connected!'))
  .catch(err => console.log('❌ MongoDB error:', err));

const appointmentSchema = new mongoose.Schema({
  userId: { type: String, required: true },
  name: String,
  phone: String,
  email: String,
  hospital: String,
  doctor: String,
  timeSlot: String,
  date: String,
  reason: String
});

const Appointment = mongoose.model('Appointment', appointmentSchema);

const userSchema = new mongoose.Schema({
  email: { type: String, required: true, unique: true },
  password: { type: String, required: true }
});
```

```

const User = mongoose.model('User', userSchema);

app.post('/api/appointments', async (req, res) => {
  try {
    const { name, phone, email, hospital, doctor, timeSlot, date,
reason } = req.body;
    const userId = req.session.userId || 'tempUserId';
    const appointment = new Appointment({ userId, name, phone,
email, hospital, doctor, timeSlot, date, reason });
    const saved = await appointment.save();
    console.log('✅ Appointment booked:', saved);
    res.status(201).json(saved);
  } catch (err) {
    console.error('❌ Error saving:', err.message);
    res.status(500).json({ error: err.message });
  }
});

app.get('/api/appointments', async (req, res) => {
  try {
    const appointments = await Appointment.find();
    res.json(appointments);
  } catch (err) {
    res.status(500).json({ error: 'Failed to fetch appointments.'
});
  }
});

app.delete('/api/appointments/:id', async (req, res) => {
  try {
    const appointment = await
Appointment.findByIdAndDelete(req.params.id);
    if (!appointment) throw new Error('Appointment not found');
    res.status(200).json({ message: 'Appointment cancelled' });
  } catch (err) {
    res.status(500).json({ error: err.message });
  }
});

app.get('/api/appointments/checkAvailability', async (req, res) => {
  try {
    const { doctor, date, timeSlot } = req.query;
    if (!doctor || !date || !timeSlot) {
      return res.status(400).json({ available: false, error:

```

```

'Missing parameters' });
    }
    const existing = await Appointment.findOne({ doctor, date,
timeSlot });
    res.json({ available: !existing });
  } catch (err) {
    res.status(500).json({ available: false, error: err.message });
  }
});

app.post('/api/signup', async (req, res) => {
  try {
    const { email, password } = req.body;
    const existingUser = await User.findOne({ email });
    if (existingUser) return res.status(400).json({ error: 'Email
already in use' });
    const user = new User({ email, password });
    await user.save();
    res.status(201).json({ message: 'User created successfully' });
  } catch (err) {
    res.status(500).json({ error: err.message });
  }
});

app.post('/api/login', async (req, res) => {
  try {
    const { email, password } = req.body;
    const user = await User.findOne({ email, password });
    if (!user) return res.status(401).json({ error: 'Invalid
credentials' });
    req.session.userId = user._id;
    res.status(200).json({ message: 'Login successful' });
  } catch (err) {
    res.status(500).json({ error: err.message });
  }
});

app.get('/api/logout', (req, res) => {
  req.session.destroy(err => {
    if (err) {
      return res.status(500).json({ error: 'Logout failed' });
    }
    res.status(200).json({ message: 'Logged out successfully' });
  });
});

```

```

});

app.get('/', (req, res) => {
    res.sendFile(path.join(__dirname, '../frontend/html/index.html'));
});

app.get('/appointment.html', (req, res) => {
    res.sendFile(path.join(__dirname,
'../frontend/html/appointment.html'));
});

app.get('/about.html', (req, res) => {
    res.sendFile(path.join(__dirname, '../frontend/html/about.html'));
});

app.get('/login.html', (req, res) => {
    res.sendFile(path.join(__dirname, '../frontend/html/login.html'));
});

app.get('/signup.html', (req, res) => {
    res.sendFile(path.join(__dirname, '../frontend/html/signup.html'));
});

app.use((req, res) => {
    res.status(404).send('404 - Page Not Found');
});

app.listen(PORT, () => {
    console.log(`🚀 Server is running at http://localhost:${PORT}`);
});

```

appointment.html

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Smart Hospital Appointment System</title>
    <link rel="stylesheet" href="/css/appointment.css">
</head>
<body>
    <nav class="navbar">
        <div class="logo">SmartHealthCare</div>
        <ul class="nav-links">
            <li><a href="/index.html">Home</a></li>

```

```

        <li><a href="/appointment.html" class="active">Book</a></li>
        <li><a href="/about.html">About</a></li>
        <li><a href="/login.html">Login</a></li>
        <li><a href="/signup.html">Sign Up</a></li>
    </ul>
</nav>

<div class="container">
    <h2>Book a Hospital Appointment</h2>
    <form id="appointmentForm">
        <label for="name">Full Name:</label>
        <input type="text" id="name" name="name" required>

        <label for="phone">Phone Number:</label>
        <input type="tel" id="phone" name="phone" required>

        <label for="email">Email:</label>
        <input type="email" id="email" name="email" required>

        <label for="hospital">Select Hospital:</label>
        <select id="hospital" name="hospital" required>
            <option value="" selected disabled>Select a
Hospital</option>
            <option value="Apollo">Apollo Hospitals</option>
            <option value="Fortis">Fortis Healthcare</option>
            <option value="Yashodha">Yashodha Hospitals</option>
            <option value="NIMS">NIMS</option>
            <option value="Rainbow">Rainbow Hospitals</option>
        </select>

        <label for="doctor">Select Doctor:</label>
        <select id="doctor" name="doctor" required disabled>
            <option value="">Select Hospital First</option>
        </select>

        <label for="timeSlot">Select Time Slot:</label>
        <select id="timeSlot" name="timeSlot" required disabled>
            <option value="">Select Doctor First</option>
        </select>

        <label for="date">Select Date:</label>
        <input type="date" id="date" name="date" required>

        <label for="reason">Reason for Visit:</label>

```

```

        <textarea id="reason" name="reason" rows="3"
required></textarea>

        <button type="submit">Book Appointment</button>
    </form>

    <a href="/api/logout" onclick="if(confirm('Are you sure?'))
window.location.href=this.href; return false;">Logout</a>

    <div id="confirmation" style="display: none;"></div>
    <button id="bookAnotherBtn" style="display: none;">Book Another
Appointment</button>
    <div id="appointmentList"></div> <!-- Added for cancel
functionality -->
</div>

<script src="/js/appointment.js"></script>
</body>
</html>

```

appointment.js

```

document.addEventListener("DOMContentLoaded", function () {
    const hospitalSelect = document.getElementById("hospital");
    const doctorSelect = document.getElementById("doctor");
    const timeSlotSelect = document.getElementById("timeSlot");
    const appointmentForm = document.getElementById("appointmentForm");
    const confirmationBox = document.getElementById("confirmation");
    const appointmentList = document.getElementById("appointmentList");

    const doctors = {
        "Apollo": ["Dr. Mehta - Cardiology", "Dr. Roy - Orthopedics", "Dr.
Sharma - General Medicine"],
        "Fortis": ["Dr. Kapoor - Neurology", "Dr. Jain - Pediatrics"],
        "Yashodha": ["Dr. Gupta - ENT", "Dr. Reddy - Dermatology"],
        "NIMS": ["Dr. Varma - Gastroenterology", "Dr. Das - Oncology"],
        "Rainbow": ["Dr. Rao - Gynecology", "Dr. Iyer - Pediatrics"]
    };

    const timeSlots = {
        "Dr. Mehta - Cardiology": ["10:00 AM", "12:00 PM", "3:00 PM"],
        "Dr. Roy - Orthopedics": ["9:30 AM", "11:00 AM", "4:30 PM"],
        "Dr. Sharma - General Medicine": ["10:15 AM", "2:00 PM", "5:30
PM"],
    };

```



```

    "Dr. Kapoor - Neurology": ["9:00 AM", "1:30 PM"],
    "Dr. Jain - Pediatrics": ["11:45 AM", "3:15 PM"],
    "Dr. Gupta - ENT": ["10:30 AM", "2:45 PM"],
    "Dr. Reddy - Dermatology": ["8:30 AM", "12:15 PM"],
    "Dr. Varma - Gastroenterology": ["9:45 AM", "1:00 PM"],
    "Dr. Das - Oncology": ["10:30 AM", "4:15 PM"],
    "Dr. Rao - Gynecology": ["9:15 AM", "3:45 PM"],
    "Dr. Iyer - Pediatrics": ["10:45 AM", "1:15 PM"]
  };

  hospitalSelect.innerHTML = '<option value="" selected disabled>Select Hospital</option>';
  Object.keys(doctors).forEach(hospital => {
    const option = document.createElement("option");
    option.value = hospital;
    option.textContent = hospital;
    hospitalSelect.appendChild(option);
  });

  async function loadAppointments() {
    appointmentList.innerHTML = "";
    try {
      const res = await fetch("/api/appointments");
      const appointments = await res.json();
      if (!appointments.length) {
        appointmentList.innerHTML = "<p>No appointments booked yet.</p>";
      } else {
        appointments.forEach(appt => {
          const div = document.createElement("div");
          div.classList.add("appointment-item");
          div.innerHTML = `
            <p><strong>${appt.name}</strong> -
            ${appt.hospital} - ${appt.doctor}</p>
            <p>${appt.date} at ${appt.timeSlot}</p>
            <button class="cancel-btn"
            data-id="${appt._id}">Cancel</button>
          `;
          appointmentList.appendChild(div);
        });
      }
    } catch (err) {
      appointmentList.innerHTML = "<p>Failed to load appointments.</p>";
    }
  }

```

```

        console.error(err);
    }
}

hospitalSelect.addEventListener("change", function () {
    const selectedHospital = this.value;
    doctorSelect.innerHTML = '<option value="">Select
Doctor</option>';
    timeSlotSelect.innerHTML = '<option value="">Select Time
Slot</option>';
    doctorSelect.disabled = true;
    timeSlotSelect.disabled = true;
    if (selectedHospital && doctors[selectedHospital]) {
        doctorSelect.disabled = false;
        doctors[selectedHospital].forEach(doc => {
            const option = document.createElement("option");
            option.value = doc;
            option.textContent = doc;
            doctorSelect.appendChild(option);
        });
    }
});

doctorSelect.addEventListener("change", function () {
    const selectedDoctor = this.value;
    timeSlotSelect.innerHTML = '<option value="">Select Time
Slot</option>';
    timeSlotSelect.disabled = true;
    if (selectedDoctor && timeSlots[selectedDoctor]) {
        timeSlotSelect.disabled = false;
        timeSlots[selectedDoctor].forEach(slot => {
            const option = document.createElement("option");
            option.value = slot;
            option.textContent = slot;
            timeSlotSelect.appendChild(option);
        });
    }
});

appointmentForm.addEventListener("submit", async function (e) {
    e.preventDefault();
    const name = document.getElementById("name").value.trim();
    const phone = document.getElementById("phone").value.trim();
    const email = document.getElementById("email").value.trim();

```

```

const hospital = hospitalSelect.value;
const doctor = doctorSelect.value;
const timeSlot = timeSlotSelect.value;
const date = document.getElementById("date").value;
const reason = document.getElementById("reason").value.trim();

if (!hospital || !doctor || !timeSlot) {
    alert("Please select a hospital, doctor, and time slot.");
    return;
}

const availabilityRes = await
fetch(`/api/appointments/checkAvailability?doctor=${encodeURIComponent(d
octor)}&date=${encodeURIComponent(date)}&timeSlot=${encodeURIComponent(t
imeSlot)}`);
const availability = await availabilityRes.json();
if (!availability.available) {
    alert("This time slot is not available. Please choose
another.");
    return;
}

const newAppointment = { userId: 'tempUserId', name, phone, email,
hospital, doctor, timeSlot, date, reason }; // Temp userId
try {
    const res = await fetch("/api/appointments", {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify(newAppointment)
    });
    if (!res.ok) throw new Error("Failed to book appointment.");
    const saved = await res.json();
    const formattedDate = new
Date(saved.date).toLocaleDateString('en-US', {
        weekday: 'long', year: 'numeric', month: 'long', day:
'numeric'
    });
    confirmationBox.innerHTML = `
        <h3>Appointment Confirmed!</h3>
        <p><strong>Name:</strong> ${saved.name}</p>
        <p><strong>Phone:</strong> ${saved.phone}</p>
        <p><strong>Email:</strong> ${saved.email}</p>
        <p><strong>Hospital:</strong> ${saved.hospital}</p>
        <p><strong>Doctor:</strong> ${saved.doctor}</p>

```

```

        <p><strong>Date:</strong> ${formattedDate}</p>
        <p><strong>Time Slot:</strong> ${saved.timeSlot}</p>
        <p><strong>Reason:</strong> ${saved.reason}</p>
        <p><strong>Queue Delay:</strong> 5 minutes</p>
    `;
    confirmationBox.style.display = "block";
    appointmentForm.reset();
    doctorSelect.disabled = true;
    timeSlotSelect.disabled = true;
    loadAppointments();
    document.getElementById("bookAnotherBtn").style.display =
"block";
    } catch (err) {
        alert("Error saving appointment.");
        console.error(err);
    }
});

appointmentList.addEventListener("click", async function (e) {
    if (e.target.classList.contains("cancel-btn")) {
        const id = e.target.dataset.id;
        try {
            await fetch(`/api/appointments/${id}`, { method: "DELETE"
});
            loadAppointments();
        } catch (err) {
            console.error("Failed to cancel appointment:", err);
        }
    }
});

loadAppointments();
});

```

signup.js

```

document.querySelector('form').addEventListener('submit', async
function(e) {
    e.preventDefault();
    const email = document.getElementById('email').value;
    const password = document.getElementById('password').value;
    try {

```

```

    const res = await fetch('/api/signup', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ email, password })
    });
    const data = await res.json();
    if (res.ok) {
      alert(data.message);
      window.location.href = '/login.html';
    } else {
      alert(data.error);
    }
  } catch (err) {
    alert('Error signing up. ');
    console.error(err);
  }
});

```

appointment.css

```

/* Google Font */
@import
url('https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap');

* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
  font-family: 'Poppins', sans-serif;
}

body {
  background: url('/assets/image.jpg') no-repeat center center fixed;
  background-size: cover;
  min-height: 100vh;
  overflow-y: auto;
  padding: 0;
  backdrop-filter: blur(15px);
}

.navbar {
  background-color: #006064;

```

```

padding: 1rem 2rem;
display: flex;
justify-content: space-between;
align-items: center;
color: white;
position: sticky;
top: 0;
z-index: 1000;
}

.navbar .logo {
  font-size: 1.5rem;
  font-weight: bold;
}

.nav-links {
  list-style: none;
  display: flex;
  gap: 20px;
}

.nav-links li a {
  color: white;
  text-decoration: none;
  font-weight: 500;
}

.nav-links li a:hover,
.nav-links li a.active {
  text-decoration: underline;
}

.container {
  width: 95%;
  max-width: 550px;
  margin: 40px auto;
  text-align: center;
  background: rgba(255, 255, 255, 0.5);
  backdrop-filter: blur(20px);
  border-radius: 12px;
  padding: 25px;
  box-shadow: 0 6px 15px rgba(0, 0, 0, 0.2);
  animation: fadeIn 0.5s ease-in-out;
}

```

```

h2 {
  color: #2c3e50;
  font-weight: 600;
  margin-bottom: 20px;
}

form {
  display: flex;
  flex-direction: column;
  gap: 15px;
}

label {
  font-weight: 600;
  text-align: left;
  color: #333;
}

input, select, textarea {
  width: 100%;
  padding: 12px;
  border-radius: 8px;
  border: 1px solid #bbb;
  font-size: 15px;
  transition: 0.3s;
  background: rgba(255, 255, 255, 0.7);
}

input:focus, select:focus, textarea:focus {
  border-color: #4ca2cd;
  box-shadow: 0px 0px 8px rgba(76, 162, 205, 0.5);
}

button {
  background: linear-gradient(135deg, #4ca2cd, #67B26F);
  color: white;
  padding: 14px;
  border: none;
  border-radius: 8px;
  font-size: 16px;
  cursor: pointer;
  transition: 0.3s;
  font-weight: 600;
}

```

```

}

button:hover {
  background: linear-gradient(135deg, #67B26F, #4ca2cd);
  transform: scale(1.05);
}

#confirmation {
  margin-top: 15px;
  padding: 12px;
  background: rgba(223, 240, 216, 0.8);
  color: #2c5e30;
  border-radius: 8px;
  font-weight: 600;
  text-align: center;
}

@keyframes fadeIn {
  from {
    opacity: 0;
    transform: translateY(-15px);
  }
  to {
    opacity: 1;
    transform: translateY(0);
  }
}

@media (max-width: 480px) {
  .container {
    width: 100%;
    padding: 20px;
  }

  .navbar {
    flex-direction: column;
    align-items: flex-start;
  }

  .nav-links {
    flex-wrap: wrap;
    gap: 10px;
    margin-top: 10px;
  }
}}

```


5. RESULTS

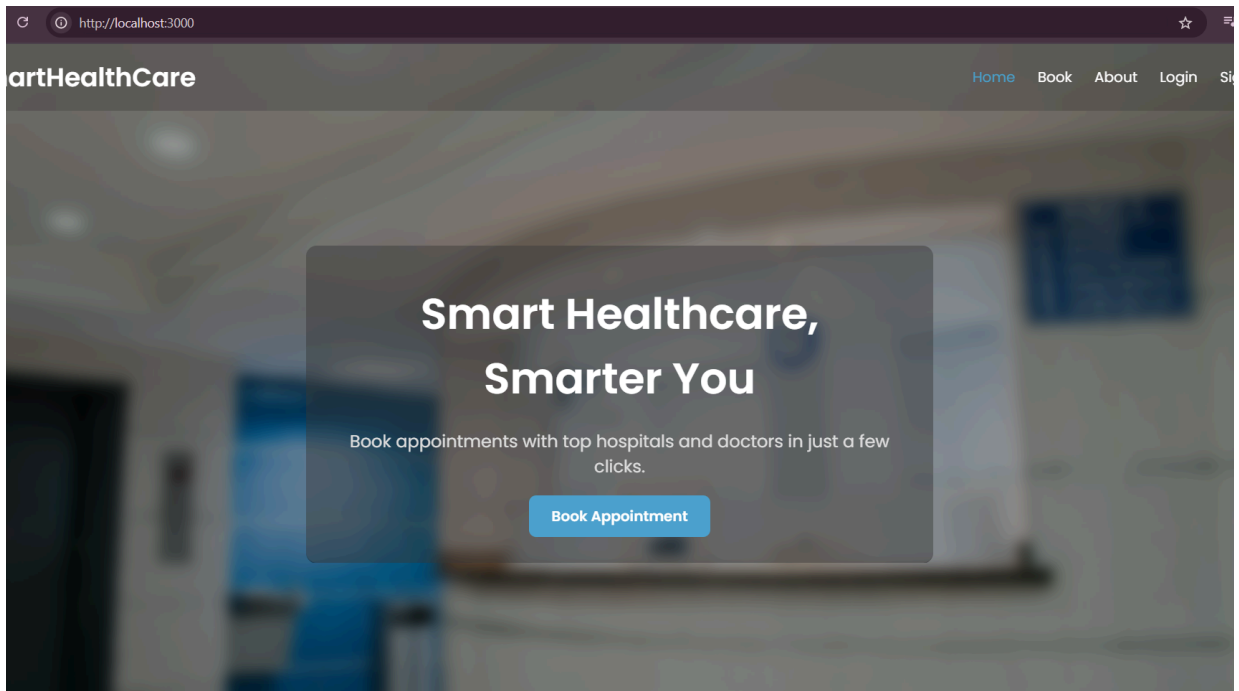


Fig 5.1 Home Page

Home page interface with a welcoming message and call-to-action to book appointments.

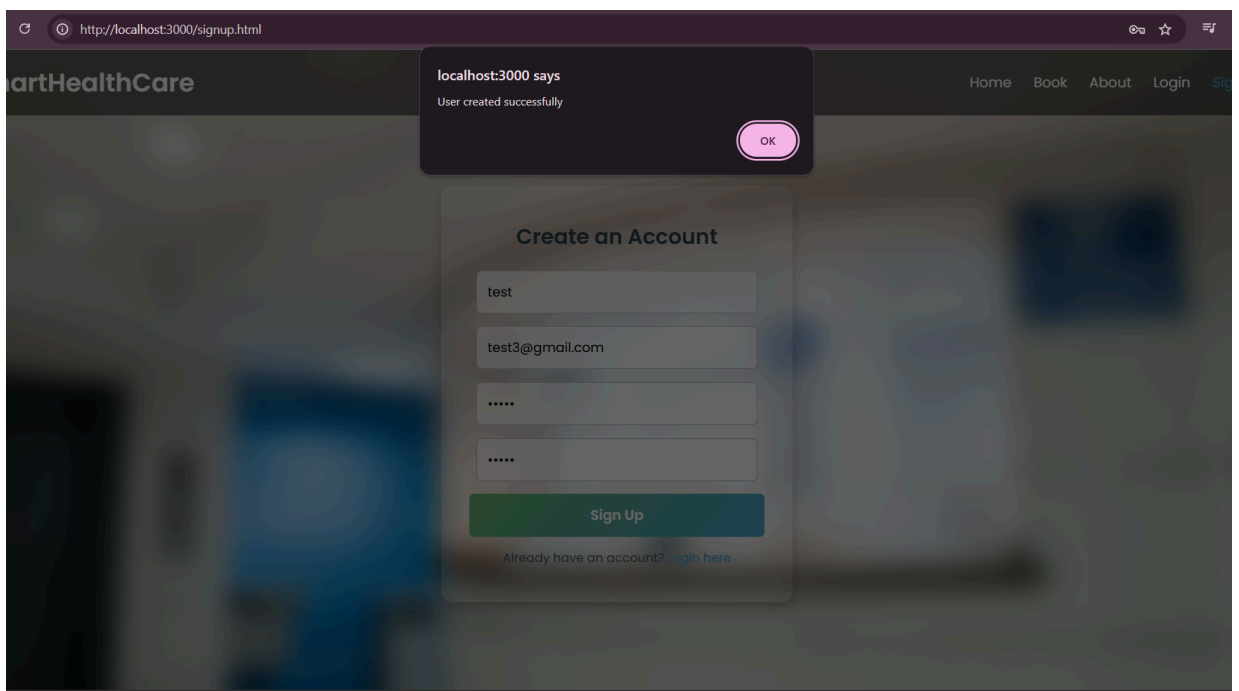


Fig 5.2 Signup Page

User signup interface with email and password fields and confirmation alert.

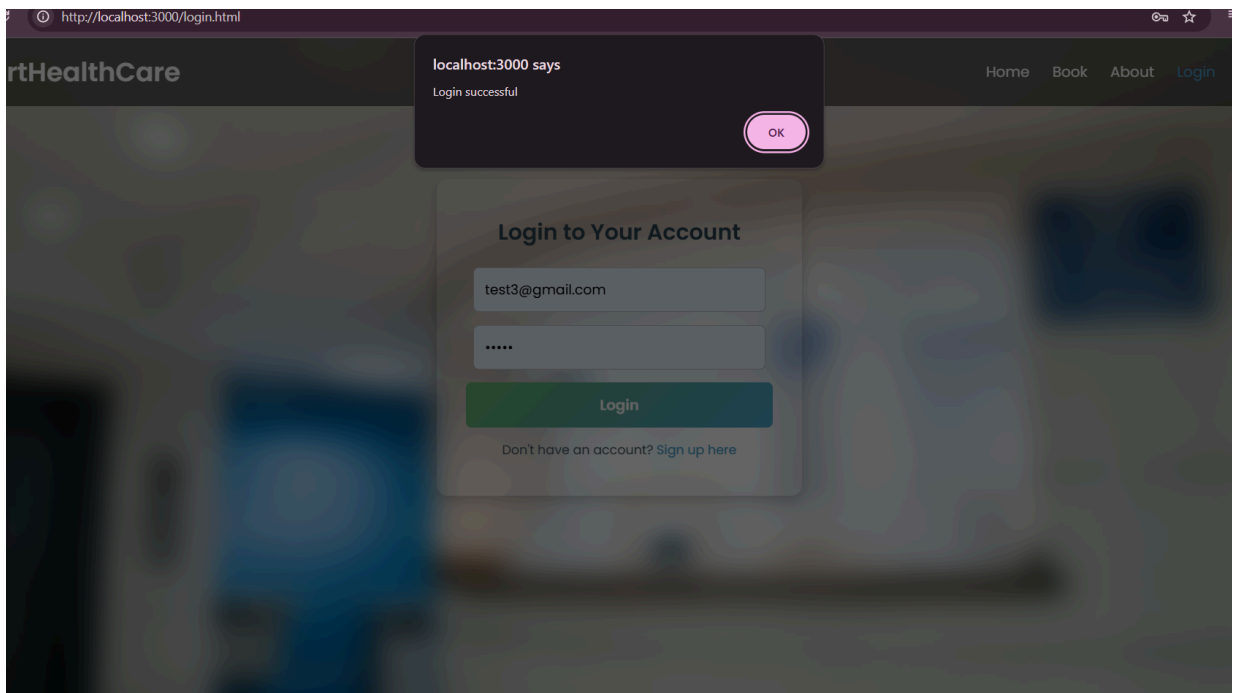


Fig 5.3 Login Page
User login interface with email and password fields and confirmation alert

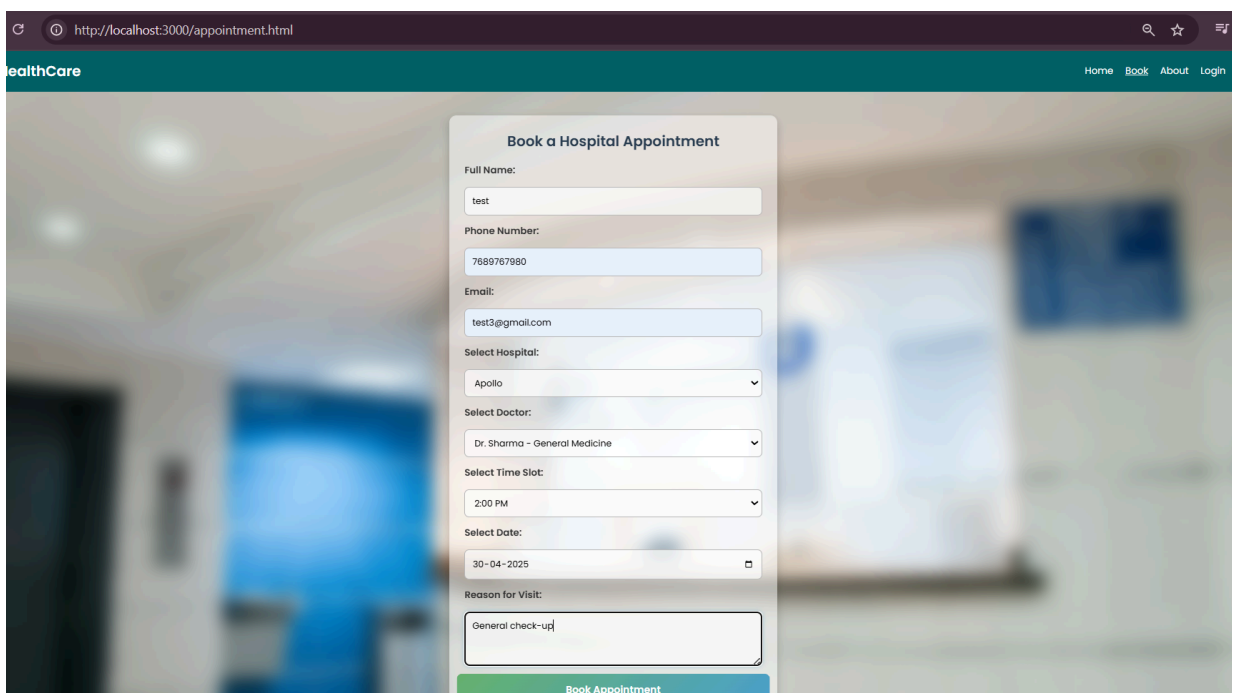


Fig 5.4 Appointment Booking Page
Appointment booking interface with form, real-time availability check, and confirmation

details

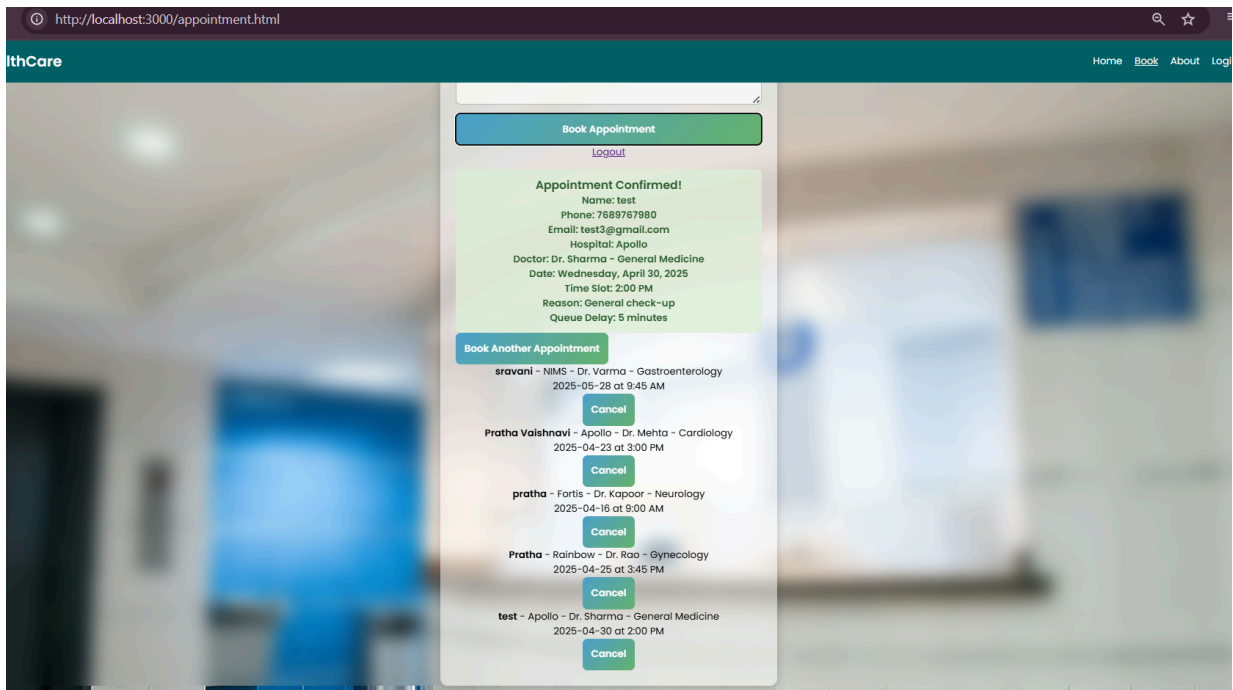
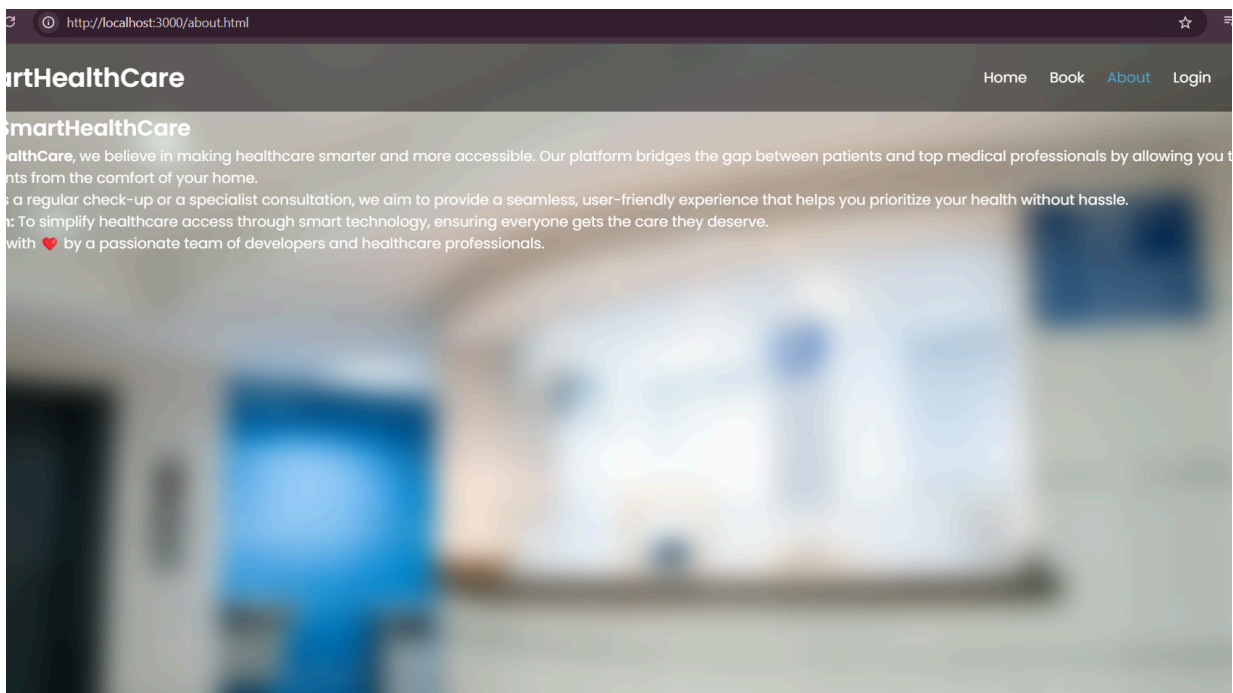


Fig 5.5 Appointment Confirmation Page

Appointment booking interface with form, real-time availability check, and confirmation details



5.6 About Page

About page detailing the system's purpose and contact information

6. CONCLUSION

This system transforms healthcare appointment management, offering efficiency, transparency, and accessibility. With a solid foundation and planned enhancements, it promises scalable impact in digital healthcare. The Smart Healthcare Appointment System presents a practical and innovative solution to the longstanding challenges faced by traditional appointment scheduling methods in healthcare settings. By digitizing and automating the booking process, the system significantly improves efficiency, reduces administrative workload, and enhances the overall patient experience. Features such as real-time queue tracking, instant notifications, and a user-friendly interface with accessibility and multilingual support ensure that the platform is inclusive, responsive, and adaptable to various user needs.

7. FUTURE SCOPE

- The Smart Healthcare Appointment System has been designed with scalability and future integration in mind, offering a strong foundation for ongoing enhancements in digital healthcare delivery. As the demand for more intelligent and patient-centered healthcare systems continues to grow, there are several opportunities to expand and improve the system.
- **AI-Powered Scheduling:** Dynamic slot adjustments based on availability.
- **Mobile App Version:** Cross-platform access for broader reach.
- **Calendar Integration:** Sync with patient calendars for convenience.
- **Hospital ERP Integration:** Seamless linkage with hospital systems.

8. REFERENCES

<https://nodejs.org/>
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<https://chatgpt.com>